# IDENTIFYING THE DERIVED INTERNAL HOM ALONG A RESOLUTION WITH A COMPOSITION OF ADJOINTS

THE STACKS PROJECT AUTHORS

*Editorial note.* All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter source sources/stacks-dualizing/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

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*Allowed standard prerequisites (editorial).*

- The adjunction between the derived internal hom of the resolution and the derived tensor product with it. Invoked at the start of the counted proof (source lines 2556-2558). Shows the functor on the resolution side is a right adjoint of the same tensor functor.
- The description of the derived internal hom along the ring map as the right adjoint of restriction. Invoked inside the counted proof (source line 2560). Identifies the functor that has to correspond under the equivalence.

*Source reference.* Original reference dga-lemma-tensor-hom-adjoint: sources/stacks-dualizing/pinned-original-source.tex, source lines 2558–2558. Navigation pointer only: the counted text refers to the Stacks item dga-lemma-tensor-hom-adjoint, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

*Source reference.* Original lemma lemma-right-adjoint: sources/stacks-dualizing/pinned-original-source.tex, source lines 2428–2428. Navigation pointer only: the counted text refers to the Stacks item lemma-right-adjoint, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

*Source reference.* Original reference situation-resolution: sources/stacks-dualizing/pinned-original-source.tex, source lines 2512–2512. Navigation pointer only: the counted text refers to the Stacks item situation-resolution, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the

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**Lemma 0.1.** *In Situation situation-resolution the functor $R\mathrm{Hom}(A, -)$ is equal to the composition of $R\mathrm{Hom}(E, -) : D(R) \to D(E, d)$ and the equivalence $- \otimes^{\mathbf{L}}_E A : D(E, d) \to D(A)$.*

*Proof.* This is true because $R\mathrm{Hom}(E, -)$ is the right adjoint to $- \otimes^{\mathbf{L}}_R E$, see Differential Graded Algebra, Lemma dga-lemma-tensor-hom-adjoint. Hence this functor plays the same role as the functor $R\mathrm{Hom}(A, -)$ for the map $R \to A$ (Lemma lemma-right-adjoint), whence these functors must correspond via the equivalence $- \otimes^{\mathbf{L}}_E A : D(E, \mathrm{d}) \to D(A)$. □